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United States Patent	12406414
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Sahoo; Rakesh Ranjan et al.

Building management system with audiovisual dynamic and interactive fault presentation interfaces

Abstract

A method for a building management system includes detecting a fault from timeseries building data, classifying a time period of the timeseries building data as a faulty time period based on the fault, and playing an audiovisual presentation of the fault and the timeseries building data by visually advancing through the timeseries building data in a chronological order and emitting a tone when the timeseries building data for the faulty time period is presented.

Inventors: **Sahoo; Rakesh Ranjan (Khordha, IN), Shelar; Mohan J. (Satara, IN), Rajagopal; Kathiresan (Thane, IN), Dsouza; Lawrence (Dombivali-East, IN), Ansari; Sk Sartaj (Purulia, IN), Sonnekar; Renuka (Parbhani, IN)**

Applicant: **Tyco Fire & Security GmbH** (Neuhausen am Rheinfall, CH)

Family ID: **89996884**

Assignee: **TYCO FIRE & SECURITY GMBH (Neuhausen am Rheinfall, CH)**

Appl. No.: **17/896420**

Filed: **August 26, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240070942 A1	Feb. 29, 2024

Publication Classification

Int. Cl.: **G06T11/20** (20060101); **G05B23/02** (20060101); **G06F3/0484** (20220101); **G06F3/16** (20060101)

U.S. Cl.:

CPC **G06T11/206** (20130101); **G05B23/0216** (20130101); **G06F3/16** (20130101); G05B2223/06 (20180801); G06F3/0484 (20130101); G06T2200/24 (20130101)

Field of Classification Search

CPC: G06T (11/206); G06F (3/16); G06F (3/0484); G05B (23/0216)

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Primary Examiner: Wu; Sing-Wai

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

BACKGROUND

(1) The present disclosure relates generally to the field of building management systems (BMSs). A BMS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof. A BMS can include tools for communicating building data and insights to a user.

(2) Operation of building equipment is largely opaque to human technicians, because, among other reasons, equipment is located in hard-to-access areas (building plenums, rooftops, behind walls, etc.) often remote from skilled technicians, various conditions/states/etc. are not sensible by a human at sufficient resolution for detailed analysis (e.g., humidity, temperature, vibrations, fan speeds, etc.), building timeseries data can include a larger number of points with values provided at a higher frequency than can be directly handled by a human in real-time, etc. Accordingly, it is

inherently technical problem to collect, arrange, and present building timeseries data relating to building equipment and/or building management systems in an understandable, accessible, and insightful manner. Inelegant presentation of mass building data can obscure relevant information, confuse expert technicians, and otherwise not address the inherent technical challenges of monitoring data for useful insights into equipment performance. Computer-machine interface design which overwhelms the user is also a technical downfall of such systems which should be avoided.

SUMMARY

(3) One implementation of the present disclosure is a method for a building management system. The method includes detecting a fault from timeseries building data, classifying a time period of the timeseries building data as a faulty time period responsive to detecting the fault, playing an audiovisual presentation of the fault and the timeseries building data by visually advancing through the timeseries building data in a chronological order such that at least a normal time period and the faulty time period are presented at different times and emitting an audio tone when the timeseries building data for the faulty time period is presented.

(4) In some embodiments, the method includes determining a quality of the tone based on a type or severity of the fault. The quality of the tone may include a pitch or volume of the tone. In some embodiments, a pitch of the tone is a function of an amount of deviation of a point of the timeseries building data from a target or threshold value for the point. In some embodiments, playing the audiovisual presentation is performed via a screen and a speaker, and wherein visually advancing the timeseries building data comprises moving a chart of the timeseries building data across the screen, wherein a width of the chart exceeds a width of the screen. In some embodiments, the method includes stopping the tone when the presentation advances such that the timeseries building data is no longer presented.

(5) Another implementation of the present disclosure is a method. The method includes determining a data loss subperiod, a fault subperiod, and a normal subperiod of a time period for a point of timeseries building data. The method also includes displaying a graph that includes a line plot of the point. The line plot has a first visual characteristic in the data loss subperiod, the line plot has a second visual characteristic in the fault subperiod, and the line plot has a third visual characteristic in the normal subperiod.

(6) In some embodiments, the method also includes hiding, in response to selection of a corresponding option by a user, the line plot for at least one of the data loss subperiod, the fault subperiod, or the normal subperiod. In some embodiments, the method also includes adding or removing, in response to selection of a corresponding option by a user, gridlines from the graph. In some embodiments, the method also includes displaying numerical percentages of the time period associated respectively with the data loss subperiod, the fault subperiod, and the normal subperiod.

(7) In some embodiments, the line plot corresponds to a first unit of building equipment and the graph also includes an additional line plot corresponding to a second unit of building equipment. In some embodiments, the method also includes displaying a date and day of week associated with the timeseries building data. In some embodiments, the method also includes calculating statistical metrics associated with the timeseries data and displaying the statistical metrics in response to a one-click request input by a user.

(8) In some embodiments, the time period corresponds to a first fault of a plurality of faults. The method can include receiving a request to navigate to a view of a second fault of the plurality of faults and changing the graph from the line plot to an additional line plot for an additional time period corresponding to a second fault of the plurality of faults. In some embodiments, the graph also includes a fault rule line indicating a threshold or target value for the point such that the line plot of the point is visually comparable to the fault rule line.

(9) Another implementation of the present disclosure is one or more non-transitory computer-readable media storing program instructions that, when executed by one or more processors, cause

the one or more processors to perform operations. The operations include detecting a fault from timeseries building data, classifying a time period of the timeseries building data as a faulty time period based on the fault, playing an audiovisual presentation of the fault and the timeseries building data by visually advancing through the timeseries building data in a chronological order and emitting a tone when the timeseries building data for the faulty time period is presented.
(10) In some embodiments, the operations also include determining a pitch of the tone based on a type or severity of the fault. The pitch of the tone may be a function of an amount of deviation of a point of the timeseries building data from a threshold or target value for the point. In some embodiments, the operations also include playing the audiovisual presentation by controlling a screen and a speaker. Visually advancing the timeseries building data can include moving a chart of the timeseries building data across the screen while a width of the chart exceeds a width of the screen. In some embodiments, the operations also include stopping the tone when the timeseries building data for the faulty time series is advanced such that the timeseries building data is no longer presented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.
(2) FIG. **1** is a drawing of a building equipped with a HVAC system, according to some embodiments.
(3) FIG. **2** is a block diagram of a waterside system that may be used in conjunction with the building of FIG. **1**, according to some embodiments.
(4) FIG. **3** is a block diagram of an airside system that may be used in conjunction with the building of FIG. **1**, according to some embodiments.
(5) FIG. **4** is a block diagram of a building management system (BMS) that may be used to monitor and/or control the building of FIG. **1**, according to some embodiments.
(6) FIG. **5** is a block diagram of another BMS which can be used to monitor and control the building of FIG. **1**, according to some embodiments.
(7) FIG. **6** is an illustration of a graphical user interface associated with a BMS, according to some embodiments.
(8) FIG. **7**A is a first frame of a storyboard-style illustration of another graphical user interface associated with a BMS, according to some embodiments.
(9) FIG. **7**B a second frame of a storyboard-style illustration of another graphical user interface associated with a BMS, according to some embodiments.
(10) FIG. **8** is a flowchart of a process for presenting building timeseries data including fault information, according to some embodiments.
DETAILED DESCRIPTION
(11) Building HVAC Systems and Building Management Systems
(12) Referring now to FIGS. **1-5**, several building management systems (BMS) and HVAC systems in which the systems and methods of the present disclosure can be implemented are shown, according to some embodiments. In brief overview, FIG. **1** shows a building **10** equipped with a HVAC system **100**. FIG. **2** is a block diagram of a waterside system **200** which can be used to serve building **10**. FIG. **3** is a block diagram of an airside system **300** which can be used to serve building **10**. FIG. **4** is a block diagram of a BMS which can be used to monitor and control building **10**. FIG.

5 is a block diagram of another BMS which can be used to monitor and control building **10**.

(13) Referring particularly to FIG. **1**, a perspective view of a building **10** is shown. Building **10** is served by a BMS. A BMS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire safety system, any other system that is capable of managing building functions or devices, or any combination thereof.

(14) The BMS that serves building **10** includes an HVAC system **100**. HVAC system **100** can include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building **10**. For example, HVAC system **100** is shown to include a waterside system **120** and an airside system **130**. Waterside system **120** can provide a heated or chilled fluid to an air handling unit of airside system **130**. Airside system **130** can use the heated or chilled fluid to heat or cool an airflow provided to building **10**. An exemplary waterside system and airside system which can be used in HVAC system **100** are described in greater detail with reference to FIGS. **2-3**.

(15) HVAC system **100** is shown to include a chiller **102**, a boiler **104**, and a rooftop air handling unit (AHU) **106**. Waterside system **120** can use boiler **104** and chiller **102** to heat or cool a working fluid (e.g., water, glycol, etc.) and can circulate the working fluid to AHU **106**. In various embodiments, the HVAC devices of waterside system **120** can be located in or around building **10** (as shown in FIG. **1**) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid can be heated in boiler **104** or cooled in chiller **102**, depending on whether heating or cooling is required in building **10**. Boiler **104** can add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller **102** can place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller **102** and/or boiler **104** can be transported to AHU **106** via piping **108**.

(16) AHU **106** can place the working fluid in a heat exchange relationship with an airflow passing through AHU **106** (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within building **10**, or a combination of both. AHU **106** can transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, AHU **106** can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid can then return to chiller **102** or boiler **104** via piping **110**.

(17) Airside system **130** can deliver the airflow supplied by AHU **106** (i.e., the supply airflow) to building **10** via air supply ducts **112** and can provide return air from building **10** to AHU **106** via air return ducts **114**. In some embodiments, airside system **130** includes multiple variable air volume (VAV) units **116**. For example, airside system **130** is shown to include a separate VAV unit **116** on each floor or zone of building **10**. VAV units **116** can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building **10**. In other embodiments, airside system **130** delivers the supply airflow into one or more zones of building **10** (e.g., via supply ducts **112**) without using intermediate VAV units **116** or other flow control elements. AHU **106** can include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU **106** can receive input from sensors located within AHU **106** and/or within the building zone and can adjust the flow rate, temperature, or other attributes of the supply airflow through AHU **106** to achieve setpoint conditions for the building zone.

(18) In FIG. **2**, waterside system **200** is shown as a central plant having a plurality of subplants **202-212**. Subplants **202-212** are shown to include a heater subplant **202**, a heat recovery chiller subplant **204**, a chiller subplant **206**, a cooling tower subplant **208**, a hot thermal energy storage (TES) subplant **210**, and a cold thermal energy storage (TES) subplant **212**. Subplants **202-212**

consume resources (e.g., water, natural gas, electricity, etc.) from utilities to serve the thermal energy loads (e.g., hot water, cold water, heating, cooling, etc.) of a building or campus. For example, heater subplant **202** may be configured to heat water in a hot water loop **214** that circulates the hot water between heater subplant **202** and building **10**. Chiller subplant **206** may be configured to chill water in a cold water loop **216** that circulates the cold water between chiller subplant **206** building **10**. Heat recovery chiller subplant **204** may be configured to transfer heat from cold water loop **216** to hot water loop **214** to provide additional heating for the hot water and additional cooling for the cold water. Condenser water loop **218** may absorb heat from the cold water in chiller subplant **206** and reject the absorbed heat in cooling tower subplant **208** or transfer the absorbed heat to hot water loop **214**. Hot TES subplant **210** and cold TES subplant **212** may store hot and cold thermal energy, respectively, for subsequent use.

(19) Hot water loop **214** and cold water loop **216** may deliver the heated and/or chilled water to air handlers located on the rooftop of building **10** (e.g., AHU **106**) or to individual floors or zones of building **10** (e.g., VAV units **116**). The air handlers push air past heat exchangers (e.g., heating coils or cooling coils) through which the water flows to provide heating or cooling for the air. The heated or cooled air may be delivered to individual zones of building **10** to serve the thermal energy loads of building **10**. The water then returns to subplants **202-212** to receive further heating or cooling.

(20) Although subplants **202-212** are shown and described as heating and cooling water for circulation to a building, it is understood that any other type of working fluid (e.g., glycol, CO2, etc.) may be used in place of or in addition to water to serve the thermal energy loads. In other embodiments, subplants **202-212** may provide heating and/or cooling directly to the building or campus without requiring an intermediate heat transfer fluid. These and other variations to waterside system **200** are within the teachings of the present invention.

(21) Each of subplants **202-212** may include a variety of equipment configured to facilitate the functions of the subplant. For example, heater subplant **202** is shown to include a plurality of heating elements **220** (e.g., boilers, electric heaters, etc.) configured to add heat to the hot water in hot water loop **214**. Heater subplant **202** is also shown to include several pumps **222** and **224** configured to circulate the hot water in hot water loop **214** and to control the flow rate of the hot water through individual heating elements **220**. Chiller subplant **206** is shown to include a plurality of chillers **232** configured to remove heat from the cold water in cold water loop **216**. Chiller subplant **206** is also shown to include several pumps **234** and **236** configured to circulate the cold water in cold water loop **216** and to control the flow rate of the cold water through individual chillers **232**.

(22) Heat recovery chiller subplant **204** is shown to include a plurality of heat recovery heat exchangers **226** (e.g., refrigeration circuits) configured to transfer heat from cold water loop **216** to hot water loop **214**. Heat recovery chiller subplant **204** is also shown to include several pumps **228** and **230** configured to circulate the hot water and/or cold water through heat recovery heat exchangers **226** and to control the flow rate of the water through individual heat recovery heat exchangers **226**. Cooling tower subplant **208** is shown to include a plurality of cooling towers **238** configured to remove heat from the condenser water in condenser water loop **218**. Cooling tower subplant **208** is also shown to include several pumps **240** configured to circulate the condenser water in condenser water loop **218** and to control the flow rate of the condenser water through individual cooling towers **238**.

(23) Hot TES subplant **210** is shown to include a hot TES tank **242** configured to store the hot water for later use. Hot TES subplant **210** may also include one or more pumps or valves configured to control the flow rate of the hot water into or out of hot TES tank **242**. Cold TES subplant **212** is shown to include cold TES tanks **244** configured to store the cold water for later use. Cold TES subplant **212** may also include one or more pumps or valves configured to control the flow rate of the cold water into or out of cold TES tanks **244**.

(24) In some embodiments, one or more of the pumps in waterside system **200** (e.g., pumps **222**,

224, **228**, **230**, **234**, **236**, and/or **240**) or pipelines in waterside system **200** include an isolation valve associated therewith. Isolation valves may be integrated with the pumps or positioned upstream or downstream of the pumps to control the fluid flows in waterside system **200**. In various embodiments, waterside system **200** may include more, fewer, or different types of devices and/or subplants Based on the particular configuration of waterside system **200** and the types of loads served by waterside system **200**.

(25) Referring now to FIG. **3**, a block diagram of an airside system **300** is shown, according to some embodiments. In various embodiments, airside system **300** may supplement or replace airside system **130** in HVAC system **100** or may be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, airside system **300** may include a subset of the HVAC devices in HVAC system **100** (e.g., AHU **106**, VAV units **116**, ducts **112-114**, fans, dampers, etc.) and may be located in or around building **10**. Airside system **300** may operate to heat or cool an airflow provided to building **10** using a heated or chilled fluid provided by waterside system **200**.

(26) In FIG. **3**, airside system **300** is shown to include an economizer-type air handling unit (AHU) **302**. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example, AHU **302** may receive return air **304** from building zone **306** via return air duct **308** and may deliver supply air **310** to building zone **306** via supply air duct **312**. In some embodiments, AHU **302** is a rooftop unit located on the roof of building **10** (e.g., AHU **106** as shown in FIG. **1**) or otherwise positioned to receive both return air **304** and outside air **314**. AHU **302** may be configured to operate exhaust air damper **316**, mixing damper **318**, and outside air damper **320** to control an amount of outside air **314** and return air **304** that combine to form supply air **310**. Any return air **304** that does not pass through mixing damper **318** may be exhausted from AHU **302** through exhaust damper **316** as exhaust air **322**.

(27) Each of dampers **316-320** may be operated by an actuator. For example, exhaust air damper **316** may be operated by actuator **324**, mixing damper **318** may be operated by actuator **326**, and outside air damper **320** may be operated by actuator **328**. Actuators **324-328** may communicate with an AHU controller **330** via a communications link **332**. Actuators **324-328** may receive control signals from AHU controller **330** and may provide feedback signals to AHU controller **330**. Feedback signals may include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by actuators **324-328**), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that may be collected, stored, or used by actuators **324-328**. AHU controller **330** may be an economizer controller configured to use one or more control algorithms (e.g., state-Based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control actuators **324-328**.

(28) Still referring to FIG. **3**, AHU **302** is shown to include a cooling coil **334**, a heating coil **336**, and a fan **338** positioned within supply air duct **312**. Fan **338** may be configured to force supply air **310** through cooling coil **334** and/or heating coil **336** and provide supply air **310** to building zone **306**. AHU controller **330** may communicate with fan **338** via communications link **340** to control a flow rate of supply air **310**. In some embodiments, AHU controller **330** controls an amount of heating or cooling applied to supply air **310** by modulating a speed of fan **338**.

(29) Cooling coil **334** may receive a chilled fluid from waterside system **200** (e.g., from cold water loop **216**) via piping **342** and may return the chilled fluid to waterside system **200** via piping **344**. Valve **346** may be positioned along piping **342** or piping **344** to control a flow rate of the chilled fluid through cooling coil **334**. In some embodiments, cooling coil **334** includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BMS controller **366**, etc.) to modulate an amount of cooling applied to supply air **310**.

(30) Heating coil **336** may receive a heated fluid from waterside system **200** (e.g., from hot water

loop **214**) via piping **348** and may return the heated fluid to waterside system **200** via piping **350**. Valve **352** may be positioned along piping **348** or piping **350** to control a flow rate of the heated fluid through heating coil **336**. In some embodiments, heating coil **336** includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BMS controller **366**, etc.) to modulate an amount of heating applied to supply air **310**.

(31) Each of valves **346** and **352** may be controlled by an actuator. For example, valve **346** may be controlled by actuator **354** and valve **352** may be controlled by actuator **356**. Actuators **354-356** may communicate with AHU controller **330** via communications links **358-360**. Actuators **354-356** may receive control signals from AHU controller **330** and may provide feedback signals to controller **330**. In some embodiments, AHU controller **330** receives a measurement of the supply air temperature from a temperature sensor **362** positioned in supply air duct **312** (e.g., downstream of cooling coil **334** and/or heating coil **336**). AHU controller **330** may also receive a measurement of the temperature of building zone **306** from a temperature sensor **364** located in building zone **306**.

(32) In some embodiments, AHU controller **330** operates valves **346** and **352** via actuators **354-356** to modulate an amount of heating or cooling provided to supply air **310** (e.g., to achieve a setpoint temperature for supply air **310** or to maintain the temperature of supply air **310** within a setpoint temperature range). The positions of valves **346** and **352** affect the amount of heating or cooling provided to supply air **310** by cooling coil **334** or heating coil **336** and may correlate with the amount of energy consumed to achieve a desired supply air temperature. AHU controller **330** may control the temperature of supply air **310** and/or building zone **306** by activating or deactivating coils **334-336**, adjusting a speed of fan **338**, or a combination of both.

(33) Still referring to FIG. **3**, airside system **300** is shown to include a building management system (BMS) controller **366** and a client device **368**. BMS controller **366** may include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that serve as system level controllers, application or data servers, head nodes, or master controllers for airside system **300**, waterside system **200**, HVAC system **100**, and/or other controllable systems that serve building **10**. BMS controller **366** may communicate with multiple downstream building systems or subsystems (e.g., HVAC system **100**, a security system, a lighting system, waterside system **200**, etc.) via a communications link **370** according to like or disparate protocols (e.g., LON, BACnet, etc.). In various embodiments, AHU controller **330** and BMS controller **366** may be separate (as shown in FIG. **3**) or integrated. In an integrated implementation, AHU controller **330** may be a software module configured for execution by a processor of BMS controller **366**.

(34) In some embodiments, AHU controller **330** receives information from BMS controller **366** (e.g., commands, setpoints, operating boundaries, etc.) and provides information to BMS controller **366** (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics, etc.). For example, AHU controller **330** may provide BMS controller **366** with temperature measurements from temperature sensors **362-364**, equipment on/off states, equipment operating capacities, and/or any other information that can be used by BMS controller **366** to monitor or control a variable state or condition within building zone **306**.

(35) Client device **368** may include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-Based computer interfaces, client-facing web services, web servers that provide pages to web clients, etc.) for controlling, viewing, or otherwise interacting with HVAC system **100**, its subsystems, and/or devices. Client device **368** may be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. Client device **368** may be a stationary terminal or a mobile device. For example, client device **368** may be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or non-mobile device. Client device **368** may communicate with BMS controller **366** and/or AHU controller **330** via communications link **372**.

(36) Referring now to FIG. **4**, a block diagram of a building management system (BMS) **400** is shown, according to some embodiments. BMS **400** may be implemented in building **10** to automatically monitor and control various building functions. BMS **400** is shown to include BMS controller **366** and a plurality of building subsystems **428**. Building subsystems **428** are shown to include a building electrical subsystem **434**, an information communication technology (ICT) subsystem **436**, a security subsystem **438**, a HVAC subsystem **440**, a lighting subsystem **442**, a lift/escalators subsystem **432**, and a fire safety subsystem **430**. In various embodiments, building subsystems **428** can include fewer, additional, or alternative subsystems. For example, building subsystems **428** may also or alternatively include a refrigeration subsystem, an advertising or signage subsystem, a cooking subsystem, a vending subsystem, a printer or copy service subsystem, or any other type of building subsystem that uses controllable equipment and/or sensors to monitor or control building **10**. In some embodiments, building subsystems **428** include waterside system **200** and/or airside system **300**, as described with reference to FIGS. **2-3**.
(37) Each of building subsystems **428** may include any number of devices, controllers, and connections for completing its individual functions and control activities. HVAC subsystem **440** may include many of the same components as HVAC system **100**, as described with reference to FIGS. **1-3**. For example, HVAC subsystem **440** may include a chiller, a boiler, any number of air handling units, economizers, field controllers, supervisory controllers, actuators, temperature sensors, and other devices for controlling the temperature, humidity, airflow, or other variable conditions within building **10**. Lighting subsystem **442** may include any number of light fixtures, ballasts, lighting sensors, dimmers, or other devices configured to controllably adjust the amount of light provided to a building space. Security subsystem **438** may include occupancy sensors, video surveillance cameras, digital video recorders, video processing servers, intrusion detection devices, access control devices and servers, or other security-related devices.
(38) Still referring to FIG. **4**, BMS controller **366** is shown to include a communications interface **407** and a BMS interface **409**. Interface **407** may facilitate communications between BMS controller **366** and external applications (e.g., monitoring and reporting applications **422**, enterprise control applications **426**, remote systems and applications **444**, applications residing on client devices **448**, etc.) for allowing user control, monitoring, and adjustment to BMS controller **366** and/or subsystems **428**. Interface **407** may also facilitate communications between BMS controller **366** and client devices **448**. BMS interface **409** may facilitate communications between BMS controller **366** and building subsystems **428** (e.g., HVAC, lighting security, lifts, power distribution, business, etc.).
(39) Interfaces **407**, **409** can be or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with building subsystems **428** or other external systems or devices. In various embodiments, communications via interfaces **407**, **409** may be direct (e.g., local wired or wireless communications) or via a communications network **446** (e.g., a WAN, the Internet, a cellular network, etc.). For example, interfaces **407**, **409** can include an Ethernet card and port for sending and receiving data via an Ethernet-Based communications link or network. In another example, interfaces **407**, **409** can include a WiFi transceiver for communicating via a wireless communications network. In another example, one or both of interfaces **407**, **409** may include cellular or mobile phone communications transceivers. In one embodiment, communications interface **407** is a power line communications interface and BMS interface **409** is an Ethernet interface. In other embodiments, both communications interface **407** and BMS interface **409** are Ethernet interfaces or are the same Ethernet interface.
(40) Still referring to FIG. **4**, BMS controller **366** is shown to include a processing circuit **404** including a processor **406** and memory **408**. Processing circuit **404** may be communicably connected to BMS interface **409** and/or communications interface **407** such that processing circuit **404** and the various components thereof can send and receive data via interfaces **407**, **409**.

Processor **406** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

(41) Memory **408** (e.g., memory, memory unit, storage device, etc.) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **408** may be or include volatile memory or non-volatile memory. Memory **408** may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to an exemplary embodiment, memory **408** is communicably connected to processor **406** via processing circuit **404** and includes computer code for executing (e.g., by processing circuit **404** and/or processor **406**) one or more processes described herein.

(42) In some embodiments, BMS controller **366** is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments BMS controller **366** may be distributed across multiple servers or computers (e.g., that can exist in distributed locations). Further, while FIG. **4** shows applications **422** and **426** as existing outside of BMS controller **366**, in some embodiments, applications **422** and **426** may be hosted within BMS controller **366** (e.g., within memory **408**).

(43) Still referring to FIG. **4**, memory **408** is shown to include an enterprise integration layer **410**, an automated measurement and validation (AM&V) layer **412**, a demand response (DR) layer **414**, a fault detection and diagnostics (FDD) layer **416**, an integrated control layer **418**, and a building subsystem integration later **420**. Layers **410-420** may be configured to receive inputs from building subsystems **428** and other data sources, determine optimal control actions for building subsystems **428** Based on the inputs, generate control signals Based on the optimal control actions, and provide the generated control signals to building subsystems **428**. The following paragraphs describe some of the general functions performed by each of layers **410-420** in BMS **400**.

(44) Enterprise integration layer **410** may be configured to serve clients or local applications with information and services to support a variety of enterprise-level applications. For example, enterprise control applications **426** may be configured to provide subsystem-spanning control to a graphical user interface (GUI) or to any number of enterprise-level business applications (e.g., accounting systems, user identification systems, etc.). Enterprise control applications **426** may also or alternatively be configured to provide configuration GUIs for configuring BMS controller **366**. In yet other embodiments, enterprise control applications **426** can work with layers **410-420** to optimize building performance (e.g., efficiency, energy use, comfort, or safety) Based on inputs received at interface **407** and/or BMS interface **409**.

(45) Building subsystem integration layer **420** may be configured to manage communications between BMS controller **366** and building subsystems **428**. For example, building subsystem integration layer **420** may receive sensor data and input signals from building subsystems **428** and provide output data and control signals to building subsystems **428**. Building subsystem integration layer **420** may also be configured to manage communications between building subsystems **428**. Building subsystem integration layer **420** translate communications (e.g., sensor data, input signals, output signals, etc.) across a plurality of multi-vendor/multi-protocol systems.

(46) Demand response layer **414** may be configured to optimize resource usage (e.g., electricity use, natural gas use, water use, etc.) and/or the monetary cost of such resource usage in response to satisfy the demand of building **10**. The optimization may be Based on time-of-use prices, curtailment signals, energy availability, or other data received from utility providers, distributed energy generation systems **424**, from energy storage **427** (e.g., hot TES **242**, cold TES **244**, etc.), or from other sources. Demand response layer **414** may receive inputs from other layers of BMS controller **366** (e.g., building subsystem integration layer **420**, integrated control layer **418**, etc.).

The inputs received from other layers may include environmental or sensor inputs such as temperature, carbon dioxide levels, relative humidity levels, air quality sensor outputs, occupancy sensor outputs, room schedules, and the like. The inputs may also include inputs such as electrical use (e.g., expressed in kWh), thermal load measurements, pricing information, projected pricing, smoothed pricing, curtailment signals from utilities, and the like.
(47) According to an exemplary embodiment, demand response layer **414** includes control logic for responding to the data and signals it receives. These responses can include communicating with the control algorithms in integrated control layer **418**, changing control strategies, changing setpoints, or activating/deactivating building equipment or subsystems in a controlled manner. Demand response layer **414** may also include control logic configured to determine when to utilize stored energy. For example, demand response layer **414** may determine to begin using energy from energy storage **427** just prior to the beginning of a peak use hour.
(48) In some embodiments, demand response layer **414** includes a control module configured to actively initiate control actions (e.g., automatically changing setpoints) which minimize energy costs Based on one or more inputs representative of or Based on demand (e.g., price, a curtailment signal, a demand level, etc.). In some embodiments, demand response layer **414** uses equipment models to determine an optimal set of control actions. The equipment models may include, for example, thermodynamic models describing the inputs, outputs, and/or functions performed by various sets of building equipment. Equipment models may represent collections of building equipment (e.g., subplants, chiller arrays, etc.) or individual devices (e.g., individual chillers, heaters, pumps, etc.).
(49) Demand response layer **414** may further include or draw upon one or more demand response policy definitions (e.g., databases, XML, files, etc.). The policy definitions may be edited or adjusted by a user (e.g., via a graphical user interface) so that the control actions initiated in response to demand inputs may be tailored for the user's application, desired comfort level, particular building equipment, or Based on other concerns. For example, the demand response policy definitions can specify which equipment may be turned on or off in response to particular demand inputs, how long a system or piece of equipment should be turned off, what setpoints can be changed, what the allowable set point adjustment range is, how long to hold a high demand setpoint before returning to a normally scheduled setpoint, how close to approach capacity limits, which equipment modes to utilize, the energy transfer rates (e.g., the maximum rate, an alarm rate, other rate boundary information, etc.) into and out of energy storage devices (e.g., thermal storage tanks, battery banks, etc.), and when to dispatch on-site generation of energy (e.g., via fuel cells, a motor generator set, etc.).
(50) Integrated control layer **418** may be configured to use the data input or output of building subsystem integration layer **420** and/or demand response later **414** to make control decisions. Due to the subsystem integration provided by building subsystem integration layer **420**, integrated control layer **418** can integrate control activities of the subsystems **428** such that the subsystems **428** behave as a single integrated super-system. In an exemplary embodiment, integrated control layer **418** includes control logic that uses inputs and outputs from a plurality of building subsystems to provide greater comfort and energy savings relative to the comfort and energy savings that separate subsystems could provide alone. For example, integrated control layer **418** may be configured to use an input from a first subsystem to make an energy-saving control decision for a second subsystem. Results of these decisions can be communicated back to building subsystem integration layer **420**.
(51) Integrated control layer **418** is shown to be logically below demand response layer **414**. Integrated control layer **418** may be configured to enhance the effectiveness of demand response layer **414** by enabling building subsystems **428** and their respective control loops to be controlled in coordination with demand response layer **414**. This configuration may advantageously reduce disruptive demand response behavior relative to conventional systems. For example, integrated

control layer **418** may be configured to assure that a demand response-driven upward adjustment to the setpoint for chilled water temperature (or another component that directly or indirectly affects temperature) does not result in an increase in fan energy (or other energy used to cool a space) that would result in greater total building energy use than was saved at the chiller.

(52) Integrated control layer **418** may be configured to provide feedback to demand response layer **414** so that demand response layer **414** checks that constraints (e.g., temperature, lighting levels, etc.) are properly maintained even while demanded load shedding is in progress. The constraints may also include setpoint or sensed boundaries relating to safety, equipment operating limits and performance, comfort, fire codes, electrical codes, energy codes, and the like. Integrated control layer **418** is also logically below fault detection and diagnostics layer **416** and automated measurement and validation layer **412**. Integrated control layer **418** may be configured to provide calculated inputs (e.g., aggregations) to these higher levels Based on outputs from more than one building subsystem.

(53) Automated measurement and validation (AM&V) layer **412** may be configured to verify that control strategies commanded by integrated control layer **418** or demand response layer **414** are working properly (e.g., using data aggregated by AM&V layer **412**, integrated control layer **418**, building subsystem integration layer **420**, FDD layer **416**, or otherwise). The calculations made by AM&V layer **412** may be based on building system energy models and/or equipment models for individual BMS devices or subsystems. For example, AM&V layer **412** may compare a model-predicted output with an actual output from building subsystems **428** to determine an accuracy of the model.

(54) Fault detection and diagnostics (FDD) layer **416** may be configured to provide on-going fault detection for building subsystems **428**, building subsystem devices (i.e., building equipment), and control algorithms used by demand response layer **414** and integrated control layer **418**. FDD layer **416** may receive data inputs from integrated control layer **418**, directly from one or more building subsystems or devices, or from another data source. FDD layer **416** may automatically diagnose and respond to detected faults. The responses to detected or diagnosed faults may include providing an alert message to a user, a maintenance scheduling system, or a control algorithm configured to attempt to repair the fault or to work-around the fault.

(55) FDD layer **416** may be configured to output a specific identification of the faulty component or cause of the fault (e.g., loose damper linkage) using detailed subsystem inputs available at building subsystem integration layer **420**. In other exemplary embodiments, FDD layer **416** is configured to provide “fault” events to integrated control layer **418** which executes control strategies and policies in response to the received fault events. According to an exemplary embodiment, FDD layer **416** (or a policy executed by an integrated control engine or business rules engine) may shut-down systems or direct control activities around faulty devices or systems to reduce energy waste, extend equipment life, or assure proper control response.

(56) FDD layer **416** may be configured to store or access a variety of different system data stores (or data points for live data). FDD layer **416** may use some content of the data stores to identify faults at the equipment level (e.g., specific chiller, specific AHU, specific terminal unit, etc.) and other content to identify faults at component or subsystem levels. For example, building subsystems **428** may generate temporal (i.e., time-series) data indicating the performance of BMS **400** and the various components thereof. The data generated by building subsystems **428** may include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These processes can be examined by FDD layer **416** to expose when the system begins to degrade in performance and alert a user to repair the fault before it becomes more severe.

(57) Referring now to FIG. **5**, a block diagram of another building management system (BMS) **500** is shown, according to some embodiments. BMS **500** can be used to monitor and control the

devices of HVAC system **100**, waterside system **200**, airside system **300**, building subsystems **428**, as well as other types of BMS devices (e.g., lighting equipment, security equipment, etc.) and/or HVAC equipment.

(58) BMS **500** provides a system architecture that facilitates automatic equipment discovery and equipment model distribution. Equipment discovery can occur on multiple levels of BMS **500** across multiple different communications busses (e.g., a system bus **554**, zone buses **556-560** and **564**, sensor/actuator bus **566**, etc.) and across multiple different communications protocols. In some embodiments, equipment discovery is accomplished using active node tables, which provide status information for devices connected to each communications bus. For example, each communications bus can be monitored for new devices by monitoring the corresponding active node table for new nodes. When a new device is detected, BMS **500** can begin interacting with the new device (e.g., sending control signals, using data from the device) without user interaction.

(59) Some devices in BMS **500** present themselves to the network using equipment models. An equipment model defines equipment object attributes, view definitions, schedules, trends, and the associated BACnet value objects (e.g., analog value, binary value, multistate value, etc.) that are used for integration with other systems. Some devices in BMS **500** store their own equipment models. Other devices in BMS **500** have equipment models stored externally (e.g., within other devices). For example, a zone coordinator **508** can store the equipment model for a bypass damper **528**. In some embodiments, zone coordinator **508** automatically creates the equipment model for bypass damper **528** or other devices on zone bus **558**. Other zone coordinators can also create equipment models for devices connected to their zone busses. The equipment model for a device can be created automatically based on the types of data points exposed by the device on the zone bus, device type, and/or other device attributes. Several examples of automatic equipment discovery and equipment model distribution are discussed in greater detail below.

(60) Still referring to FIG. **5**, BMS **500** is shown to include a system manager **502**; several zone coordinators **506**, **508**, **510** and **518**; and several zone controllers **524**, **530**, **532**, **536**, **548**, and **550**. System manager **502** can monitor data points in BMS **500** and report monitored variables to various monitoring and/or control applications. System manager **502** can communicate with client devices **504** (e.g., user devices, desktop computers, laptop computers, mobile devices, etc.) via a data communications link **574** (e.g., BACnet IP, Ethernet, wired or wireless communications, etc.). System manager **502** can provide a user interface to client devices **504** via data communications link **574**. The user interface may allow users to monitor and/or control BMS **500** via client devices **504**.

(61) In some embodiments, system manager **502** is connected with zone coordinators **506-510** and **518** via a system bus **554**. System manager **502** can be configured to communicate with zone coordinators **506-510** and **518** via system bus **554** using a master-slave token passing (MSTP) protocol or any other communications protocol. System bus **554** can also connect system manager **502** with other devices such as a constant volume (CV) rooftop unit (RTU) **512**, an input/output module (IOM) **514**, a thermostat controller **516** (e.g., a TEC5000 series thermostat controller), and a network automation engine (NAE) or third-party controller **520**. RTU **512** can be configured to communicate directly with system manager **502** and can be connected directly to system bus **554**. Other RTUs can communicate with system manager **502** via an intermediate device. For example, a wired input **562** can connect a third-party RTU **542** to thermostat controller **516**, which connects to system bus **554**.

(62) System manager **502** can provide a user interface for any device containing an equipment model. Devices such as zone coordinators **506-510** and **518** and thermostat controller **516** can provide their equipment models to system manager **502** via system bus **554**. In some embodiments, system manager **502** automatically creates equipment models for connected devices that do not contain an equipment model (e.g., IOM **514**, third party controller **520**, etc.). For example, system manager **502** can create an equipment model for any device that responds to a device tree request.

The equipment models created by system manager **502** can be stored within system manager **502**. System manager **502** can then provide a user interface for devices that do not contain their own equipment models using the equipment models created by system manager **502**. In some embodiments, system manager **502** stores a view definition for each type of equipment connected via system bus **554** and uses the stored view definition to generate a user interface for the equipment.

(63) Each zone coordinator **506-510** and **518** can be connected with one or more of zone controllers **524**, **530-532**, **536**, and **548-550** via zone buses **556**, **558**, **560**, and **564**. Zone coordinators **506-510** and **518** can communicate with zone controllers **524**, **530-532**, **536**, and **548-550** via zone busses **556-560** and **564** using a MSTP protocol or any other communications protocol. Zone busses **556-560** and **564** can also connect zone coordinators **506-510** and **518** with other types of devices such as variable air volume (VAV) RTUs **522** and **540**, changeover bypass (COBP) RTUs **526** and **552**, bypass dampers **528** and **546**, and PEAK controllers **534** and **544**.

(64) Zone coordinators **506-510** and **518** can be configured to monitor and command various zoning systems. In some embodiments, each zone coordinator **506-510** and **518** monitors and commands a separate zoning system and is connected to the zoning system via a separate zone bus. For example, zone coordinator **506** can be connected to VAV RTU **522** and zone controller **524** via zone bus **556**. Zone coordinator **508** can be connected to COBP RTU **526**, bypass damper **528**, COBP zone controller **530**, and VAV zone controller **532** via zone bus **558**. Zone coordinator **510** can be connected to PEAK controller **534** and VAV zone controller **536** via zone bus **560**. Zone coordinator **518** can be connected to PEAK controller **544**, bypass damper **546**, COBP zone controller **548**, and VAV zone controller **550** via zone bus **564**.

(65) A single model of zone coordinator **506-510** and **518** can be configured to handle multiple different types of zoning systems (e.g., a VAV zoning system, a COBP zoning system, etc.). Each zoning system can include a RTU, one or more zone controllers, and/or a bypass damper. For example, zone coordinators **506** and **510** are shown as Verasys VAV engines (VVEs) connected to VAV RTUs **522** and **540**, respectively. Zone coordinator **506** is connected directly to VAV RTU **522** via zone bus **556**, whereas zone coordinator **510** is connected to a third-party VAV RTU **540** via a wired input **568** provided to PEAK controller **534**. Zone coordinators **508** and **518** are shown as Verasys COBP engines (VCEs) connected to COBP RTUs **526** and **552**, respectively. Zone coordinator **508** is connected directly to COBP RTU **526** via zone bus **558**, whereas zone coordinator **518** is connected to a third-party COBP RTU **552** via a wired input **570** provided to PEAK controller **544**.

(66) Zone controllers **524**, **530-532**, **536**, and **548-550** can communicate with individual BMS devices (e.g., sensors, actuators, etc.) via sensor/actuator (SA) busses. For example, VAV zone controller **536** is shown connected to networked sensors **538** via SA bus **566**. Zone controller **536** can communicate with networked sensors **538** using a MSTP protocol or any other communications protocol. Although only one SA bus **566** is shown in FIG. **5**, it should be understood that each zone controller **524**, **530-532**, **536**, and **548-550** can be connected to a different SA bus. Each SA bus can connect a zone controller with various sensors (e.g., temperature sensors, humidity sensors, pressure sensors, light sensors, occupancy sensors, etc.), actuators (e.g., damper actuators, valve actuators, etc.) and/or other types of controllable equipment (e.g., chillers, heaters, fans, pumps, etc.).

(67) Each zone controller **524**, **530-532**, **536**, and **548-550** can be configured to monitor and control a different building zone. Zone controllers **524**, **530-532**, **536**, and **548-550** can use the inputs and outputs provided via their SA busses to monitor and control various building zones. For example, a zone controller **536** can use a temperature input received from networked sensors **538** via SA bus **566** (e.g., a measured temperature of a building zone) as feedback in a temperature control algorithm. Zone controllers **524**, **530-532**, **536**, and **548-550** can use various types of control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms,

proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control a variable state or condition (e.g., temperature, humidity, airflow, lighting, etc.) in or around building **10**.

(68) BMS with Audiovisual Dynamic and Interactive Fault Presentation Interfaces

(69) Operation of building equipment is largely opaque to human technicians, because, among other reasons, equipment is located in hard-to-access areas (building plenums, rooftops, behind walls, etc.) often remote from skilled technicians, various conditions/states/etc. are not sensible by a human at sufficient resolution for detailed analysis (e.g., humidity, temperature, vibrations, fan speeds, etc.), building timeseries data can include a larger number of points with values provided at a higher frequency than can be directly handled by a human in real-time, etc. Accordingly, it is inherently technical problem to collect, arrange, and present building timeseries data relating to building equipment and/or building management systems in an understandable, accessible, and insightful manner. Inelegant presentation of mass building data can obscure relevant information, confuse expert technicians, and otherwise not address the inherent technical challenges of monitoring data for useful insights into equipment performance. Computer-machine interface design which overwhelms the user is also a technical downfall of such systems which should be avoided. The examples of FIGS. **6-7** provide technical solutions to such inherent challenges of monitoring building equipment, including by transforming building data into a form having real, valuable utility as will be clear from the Figures and the following description.

(70) Referring now to FIG. **6**, a graphical user interface **600** is shown, according to some embodiments. The graphical user interface **600** can be presented on a client device (e.g., client device **368**, client device **448**, client device **504**) and can be generated by a building management system, for example by BMS controller **366** (e.g., by fault detection and diagnostics layer **416**), by monitoring and reporting applications **422**, and/or by system manager **502** in various embodiments. In some embodiments, the graphical user interface **600** is generated by a cloud-based computing resource and is accessible via the Internet (e.g., via an Internet browser on a client device).

(71) The graphical user interface **600** is configured to display building timeseries data for multiple points over a time period corresponding to a detected fault. In the example shown, the fault is labeled as “Supply Air Temperature is higher than set point in cooling mode.” The graphical user interface **600** can be adapted to visualize building time series data corresponding to various other types of fault, for example by navigating to another fault using arrows **602**. A header **604** identifies the relevant unit of equipment for which the fault occurred and for which data is shown, the number of occurrences of the fault in the view shown, the number of fault hours (e.g., duration of time in a fault state), and a number of similarly faulted equipment (e.g., a count of other units of equipment with the same type of fault). The number of similarly faulted equipment may be provided as an interactive link to detailed views, further information, etc. related to the similarly-faulted equipment which may cause the graphical user interface **600** to navigate to another view displaying such information when clicked or selected. The displayed fault can be saved as a favorite (bookmarked, etc.) by selecting the favorites button **606**, allowing quick navigation back to the displayed view at a later time.

(72) Selection of a comment button **608** causes the graphical user interface **600** to provide a window, field, etc. allowing free-text entry of comments by a user, for example to allow the user to provide comments on the displayed fault. The comments can be saved and later displayed by a subsequent selection of the comment button **608**.

(73) The graphical user interface **600** is shown as including multiple graphs of building timeseries data, shows as a first graph **610**, a second graph **612**, and a third graph **614**. The first graph **610**, the second graph **612**, and the third graph **614** may illustrate different points (e.g., measured variables, setpoints, statuses, etc.) of building timeseries data for a same time period. The first graph **610**, the second graph **612**, and the third graph **614** are aligned with a shared horizontal axis showing the

times corresponding to the data plotted on the graphs **610-614**. The horizontal axis may be labelled with hours, dates, and day of week. Easy comparisons, connections, etc. can thus be made across the graphs **610-614** and across days (e.g., with day of week information distinguishing week days and weekends, etc.). Other day-typing information can be included in various embodiments (e.g., business day, remote work day, holiday, weekend, school day, etc.). Each of the graphs **610-614** may have a different vertical axis, shown as on/off in the first graph **610**, ° C. in the second graph **612**, and % in the third graph **614**, corresponding to the unit of measure of the points displayed in each of the graphs **610-614**. Different points that have the same unit of measure can be presented within the same graph (e.g., one of the graphs **610-614**), whereas different points that have different units of measure can be presented in different graphs.

(74) The first graph **610** is shown as displaying runtime (on/off) data for an air handling unit. Blocks **616** represent times at which the air handling unit is off. Blocks **618** represent times at which the air handling unit is on. The blocks **618** are coded according to key **620** to indicate times at which the equipment is operating normally (periods **622**, for example shown in green), operating in a fault state (periods **624**, for example shown in red), and for which data is not available (period **626**, shown as broken/dashed illustrating data loss). A person viewing the first graph **610** can then easily see when the equipment was off, when the equipment was running, and the fault and/or data loss status of the equipment across the displayed time period.

(75) The second graph **612** is shown as including a first line plot **628** and a second line plot **630**. The first line plot **628** visualizes measurements of a physical condition of a building or equipment unit, shown in the example of FIG. **6** as a supply air temperature of an air handling unit. The second line plot **630** visualizes a setpoint for the condition plotted by the first line plot **628**. In the example of FIG. **6**, the second line plot **630** shows a supply air temperature setpoint for the air handling unit. The second graph **612** thus allows a user to easily observe how well the setpoint tracked the setpoint over a time period.

(76) The first line plot **628** and the second line plot **630** are coded by color, line type, dashing, etc. to show normal operating periods, fault periods (shown as periods **624**), and data loss periods (shown as periods **626**). The fault periods may be colored differently (e.g., red) compared to the normal periods (e.g., blue, green), and the data loss periods may be dashed, dotted, etc. As shown in FIG. **6**, for the data loss period **626** the first line plot **628** is estimated (interpolated, etc.) as a straight line between the last point before data loss and the first point after data loss (once data becomes available again), with the linear data loss periods **626** included in the graph. The second graph **612** thereby directly illustrates, in a clear manner, both point values and status information.

(77) The third graph **614** includes a third line plot **632**, a fourth line plot **634**, and a fifth line plot **636**, having similar color-coding and/or dashing as for the second graph **612** to illustrate fault periods **624** and data loss periods **626** as distinct from one another and from normal operating periods. Third graph **614** has a percentage (e.g., 0% to 100%) as the unit on the vertical axis, with the third line plot **632**, the fourth line plot **634**, and the fifth line plot **636** plotting equipment operating points as percentage values. In the example shown, the third line plot **632** plots the open/close position of a first valve or damper, the fourth line plot **634** plots the open/close position of a second valve or damper, and the fifth line plot **636** plots a fan speed. Other values can be plotted in various embodiments, for example as may be suitable to visualize performance of various types of equipment. The third graph **614** demonstrates that multiple points which have the same units can be grouped on the same graph for comparison and space-savings.

(78) The key **620** of the graphical user interface **600** is shown as indicating the colors or other characteristics used in the graphs **610**, **612**, **614**. The key **620** also includes check boxes **638**, which can be selected/deselected by a user in order to show/hide data for the corresponding category. As shown, three check boxes **638** are selected and data for three categories (data loss, fault, normal) is shown in the graphs **610**, **612**, **614**. When one or more of the check boxes is deselected, the graphs **610**, **612**, **614** will update to hide (not show) data for the corresponding category or categories. The

graphical user interface **600** thereby can be adjusted on command to hide data loss segments, hide faulty data, and/or hide normal data. The key **620** is further shown as showing percentages **640** for the categories indicating the percentages of the total time period displayed associated with each of the data loss period(s), fault period(s), and normal period(s).

(79) The graphical user interface **600** is also shown as including an add/delete points button **642**. The add/delete points button **642** is selectable to provide options to add/delete points form the graphs **610**, **612**, **614**, thereby enabling user customization of the displayed points. One or more of the graphs **610**, **612**, **614** can be automatically updated in response to a user request to add or delete a point.

(80) The graphical user interface **600** is also shown as including a statistics button **644**. In response to selection of the statistics button **644**, statistical analysis of the building timeseries data can be displayed. For example, display of statistical analysis can include flagging anomalies in fault and normal data, showing a minimum value of a variable, showing a maximum value of a variable, showing 25% and 75% quartiles, showing a percentage of data loss, etc. Such statistical analyses are advantageously automatically performed for easily-accessible display directly in the graphical user interface **600**.

(81) The graphical user interface **600** is also shown as including a grid button **646**. The grid button **646** is selectable to show or hide grid lines of the graphs **610**, **612**, **614**. Grid lines show even steps of units for both the horizontal and vertical axes across the whole graph area to improve the ability of a viewer to assess plotted values throughout the graphs **610**, **612**, **614**. Grid lines can include horizontal and vertical lines. As shown, for example, vertical lines can be provided for each 3-hour increment along the horizontal axis. Horizontal lines can also be provided, for example in increments of 25% or 50% (for plotted percentages), one degree or five degrees (for plotted temperature), etc. Grid lines can also be hidden by selection of the grid button **646** if a more open view is preferred by a user.

(82) The graphical user interface **600** also includes an export button **648**. The export button allows the user to export the graphical user interface **600** (or some reformatted version thereof), for example as an image (e.g., .png file), .xml file, .pdf file, and/or other format. The exported version retains the coloring, dashing, etc. of the graphs **610**, **612**, **614** as shown. Exporting to various file formats may be enable users to save and share results in a manner convenient to them and or convenient to other persons who may not have direct access to the graphical user interface **600**.

(83) The graphical user interface **600** is shown to include several time range selection buttons **650** which can be selected to adjust the range of time shown in the graphs **610-614**. For example, the time range selection buttons **650** are shown to include buttons labeled “WK” (one week), “1M” (one month), “3M” (three months), “6M” (six months), “1Y” (one year), and “C” (custom) which indicate different ranges of time which can be represented in the graphs **610-614**. A user can select any of the time range selection buttons **650** to cause the graphical user interface **600** to display a longer or shorter period of the timeseries data (e.g., changing the duration of the time window shown in the graphs **610-614** to effectively “zoom in” or “zoom out” on the timeseries data) or move the time window forward or backward in time (e.g., retain the same duration of the time window but slide the time window toward the past or future to show a different portion of the timeseries data). In some embodiments, selecting the custom time range button “C” may allow a user to define a custom time period via the custom time period input fields **652** by defining the beginning time of the custom time period (shown as “23 Sep 19” in FIG. **6**), the end time of the custom time period (shown as “25 Sep 19” in FIG. **6**), and/or the duration of the custom time period (shown as “72 Hrs” in FIG. **6**). Entering a custom time range via the custom time period input fields **652** may cause the graphical user interface **600** to move the time window forward or backward in time and/or expand or contract the duration of the time window shown in the graphs **610-614** to correspond to the custom time range.

(84) Referring now to FIGS. **7**A-B, a storyboard-style illustration of a graphical user interface **700**

is shown, according to some embodiments. The graphical user interface **700** can be presented on a client device (e.g., client device **368**, client device **448**, client device **504**) and can be generated by a building management system, for example by BMS controller **366** (e.g., by fault detection and diagnostics layer **416**), by monitoring and reporting applications **422**, and/or by system manager **502** in various embodiments. In some embodiments, the graphical user interface **700** is generated by a cloud-based computing resource and is accessible via the Internet (e.g., via an Internet browser on a client device).

(85) FIG. **7**A includes a first frame **701** and FIG. **7**B includes a second frame **702**, showing the graphical user interface **700** at different points at time, for example with the second frame **702** showing a view which is provided subsequent to the view shown in the first frame **701**. FIGS. **7**A-B illustrate that the graphical user interface **700** provides a dynamic audiovisual presentation in which visualizations of building timeseries data play as a presentation, video, etc. and which may include audio cues (e.g., sound, alarms, noise, tones, etc.) indicating points in time associated with faults or other events. Such an approach allows for surfacing of building timeseries data to a user without requiring the user to repeatedly navigate, zoom in, zoom out, etc. through interface views as would otherwise be required to view a comparable amount of building performance information.

(86) Advantageously, the graphical user interface **700** can “play” a set of timeseries data in a manner similar to how a media player plays an audio file or a video file. For example, in the context of a media player, a video file may include a video track which contains the visual data used to generate the visual images which are presented to the user as well as an audio track which contains the audio data used to generate the audio presented to the user. The video track may include a sequence of video frames, each of which contains the video data to be presented at a given time to form the image presented to the user. The audio track may include a sequence of audio samples which define the loudness information at a particular point in time. A media player may play both the video track and the audio track concurrently and in synchronization with each other (i.e., playing or presenting each frame of the video track and each sample of the audio track in sequence) to generate the audiovisual presentation. While the complete video file may be predefined and/or preloaded into memory prior to beginning playback, the media player advances in time through the temporally arranged video frames and audio samples to present each video frame and audio sample in a particular sequence defined by the video file.

(87) The graphical user interface **700** can play a set of timeseries data by generating an audiovisual presentation based on the values of the individual timeseries that are included. Each timeseries represented in the graphical user interface **700** can be conceptualized as a separate “track” of the timeseries data, similar to how a video file contains separate audio and video tracks. Each time step of the timeseries can be conceptualized as a separate “frame” or “sample” of the track, similar to the video frames of a video file or audio samples of an audio file. The data values of each timeseries at each time step define how that particular timeseries is rendered or presented at each time step shown in the graphical user interface **700**. As shown in FIGS. **7**A-B, the graphical user interface **700** can present multiple timeseries concurrently by plotting the timeseries data on the same graph or separate graphs which are temporally aligned along their respective time axes (shown as the horizontal axes in FIGS. **7**A-B). Different timeseries can be presented or rendered in different ways. For example, some timeseries (e.g., floating point timeseries, numerical timeseries) can be presented as line graphs in which the value of the timeseries at a given time step defines the position of the line on the graph, whereas other timeseries (e.g., binary on/off, fault yes/no, data loss yes/no) can be presented as colored bars that indicate binary values. In some embodiments, some timeseries can be presented as audio data in which a particular sound is emitted corresponding to the value of the timeseries as that sample of the timeseries is played.

(88) In some embodiments, the entire set of timeseries data used to generate the graphical user interface **700** exists prior to playing the timeseries data. For example, the timeseries may be historical data timeseries stored in one or more databases and can be retrieved or downloaded into

memory prior to starting playback. In other embodiments, the timeseries data may be partially complete prior to beginning playback and may be streamed into the graphical user interface **700** during playback (e.g., as the data are collected or provided). The graphical user interface **700** can play the timeseries data by presenting portions of the timeseries data in a predefined temporal order corresponding to the time stamps of the timeseries data, similar to how a media player plays a video file by presenting the video frames and audio samples in a predefined temporal order. However, unlike a media player which presents only a single video frame or audio sample at a given time, the graphical user interface **700** may present multiple time steps of the timeseries data concurrently. For example, the graphical user interface **700** may present a window of the timeseries data which includes multiple time steps concurrently. The window may have a specific duration (e.g., one hour, one day, one week, one month, etc.) which can be adjusted via user-facing controls, similar to the time range selection buttons **650** of the graphical user interface **600**. The window may continuously slide forward in time during playback, or can jump forward or backward in time when the user interacts with the playback controls **712** (e.g., shift the window forward or backward by one week, one hour, one day, etc.).

(89) During normal playback, the portion of the timeseries data presented to the user may correspond to the portion of the timeseries data that have timestamps within the window. In other words, the graphical user interface **700** may present the timeseries data from the perspective or reference frame of the window as the window slides forward in time. Accordingly, the samples of the timeseries data for a particular time step shown via the graphical user interface **700** may move into and across the window in a direction that corresponds to the time axis of the various graphs (e.g., from right to left in FIGS. 7A-B) such that the timeseries data appear to be sliding across the graphical user interface **700** during normal playback. In addition to the visual presentation of the timeseries data, the graphical user interface **700** may include a synchronized audio presentation of one or more of the timeseries in which a particular audible tone is played when the timestamp of the corresponding sample of the timeseries data reaches a particular location in the window. These and other features of the graphical user interface **700** are described in greater detail below.

(90) As shown in FIGS. 7A-B, the graphical user interface includes a first graph **704**, a second graph **706**, and a third graph **708**. In the example shown, the first graph **704** plots a first point of building timeseries data over time (shown as a secondary chilled water pump command) and the second graph **706** plots a second point of building timeseries data over time (shown as a secondary chilled water pump status). Historical data is displayed in the first graph **704** and the second graph **706**, i.e., corresponding to times which have already elapsed, while predicted values for future times can be displayed in other embodiments. As shown, values are plotted at multiple time steps in a displayed time period (e.g., every 15 minutes, every 30 minutes). Due to the length of the time period for which data is available and the space used to display each value, the first graph **704** and the second graph **706** have a width (left-to-right in the example shown) which exceeds a width of the graphical user interface **700** (e.g., a width of a screen of a client device displaying the graphical user interface **700**). That is, the first graph **704** and the second graph **706** extend virtually beyond the right and/or left edges of the graphical user interface **700**. The first graph **704** and the second graph **706** can be fully represented and populated in video tracks which play over time such that the full graphs are presented bay an end of a presentation.

(91) As illustrated by arrow **711**, the first graph **704** and the second graph **706** play as a visual presentation (e.g., video) in which the first graph **704** and the second graph **706** slide right-to-left such that the displayed time period (and associated data) advances in time. Time steps plotted off of the right of the graphical user interface **700** come into view and move toward the left of the screen, with a comparable number of time steps disappear from view off the left edge of the graphical user interface **700**. For example, the first graph **704** and the second graph **706** can advance from the view shown in the first frame **701** to the view shown in the second frame **702**, where data for relatively later time steps is shown. This advancement of the first graph **704** and the second graph

706 across the graphical user interface **700** allows more data to be seen by a user as compared to a static screen, without requiring the user to navigate, zoom in, zoom out, etc. The first graph **704** and the second graph **706** can move at a same speed to maintain alignment of the horizontal axis (shown as time steps) of the first graph **704** and the second graph **706**.

(92) As shown in FIGS. **7**A-B, the first graph **704** and the second graph **706** are shaded (colored, highlighted, etc.) to visually distinguish periods of normal operation from periods of faulty operation (e.g., corresponding to occurrence of one or more faults). As shown in the example, normal periods have a white background while faulty periods are marked with a background coloration or pattern (e.g., in period **710**). Such shading, coloration, pattern, etc. moves with other portions of the first graph **704** and the second graph **706** as the visual presentation advances across the graphical user interface **700**. In some embodiments, the first graph **704**, the second graph **706**, and the third graph **708** of FIGS. **7**A-B are configured in the same or similar manner as the first graph **610**, the second graph **612**, and/or the third graph **614** of FIG. **6** and described above.

(93) In some embodiments, audio is provided with the visual presentation, for example by causing a speaker of a client device to emit sound based on one or more aspects of the building timeseries data. In some embodiments, a tone is emitted while a fault period (e.g., period **710**) is crossing a point or segment of the graphical user interface **700** (e.g., a vertical center line of the graphical user interface **700**). An audible cue is thus provided to a user to inform the user that a fault period is being displayed. Such an audible cue may be easier for a user to notice or recognize, especially when the user's attention is not fully on the graphical user interface **700**. In some embodiments, a characteristic of the tone emitted is based on a characteristic of the fault. For example, a pitch, duration, volume, timbre, etc. of the sound may vary based on a severity, criticality, degree, type, etc. of a fault. As one example, the pitch and/or volume may increase as a deviation of a measured value from a setpoint or threshold value (e.g., alarm limit) increases. The audio tone may thereby communicate information to a user which is not visually displayed and/or hard to interpret from the visual display. The audio tone may also facilitate use of the graphical user interface **700** by visually-impaired users. The audio tone may sound once at a beginning of display of a fault period or may continue as long as the fault period is moving across a particular part of the graphical user interface **700** and stop once the fault period has moved passed such a point.

(94) The third graph **708** may also be play (advance, etc.) as a visual presentation (e.g., video). The third graph **708** can update by plotting more (e.g., new, updated) values of the graphed variable over time, for example as new data is collected and/or as the time presented by the audio/visual presentation advances. The time span shown on the third graph **708** may be static, with the plotted values updating over time. Accordingly, the second frame **702** shows more values plotted in the third graph **708** as compared to the first frame **701**. In some embodiments, a track of the presentation for the third graph **708** is synced with one or more tracks of the presentation for the first graph **704** and/or the second graph **706**, such that the added values are coordinated with an aspect of the change in the first graph **704** and/or the second graph **706** as the presentation plays. Dynamic presentation of the variable plotted in the third graph **708** is thereby provided.

(95) The graphical user interface **700** is shown as including playback controls **712**, which include buttons to stop/start playing of the audiovisual presentation (advancement of the graphs), jump back in time along the graphs by a week, a day, or an hour, and to jump forward in time along the graphs by a week, a day, or an hour. A user can thus easily find and/or review segments of the audiovisual presentation of particular interest to a user.

(96) The graphical user interface **700** is also shown as including a data aggregation option **714**, which provides options relating to the frequency of data displayed on the graph (allowing for longer or shorter timescales and/or higher or lower frequencies to be displayed). Various such customizability can be provided in various embodiments.

(97) Referring now to FIG. **8**, a flowchart of a process **800** for generating presentations of building timeseries data including fault information is shown, according to some embodiments. Process **800**

can output the graphical user interface **600** of FIG. **6** and/or the graphical user interface **700** of FIGS. **7**A-B in some embodiments. Process **800** can be executed by processing circuitry programmed to perform the operations of process **800**, for example one or more processors in combination with one or more non-transitory computer memory storing instructions which, when executed by the one or more processors, causes the one or more processors to perform the operations of process **800**. Such processing circuitry can be included in a building management system, for example in BMS controller **366** (e.g., using fault detection and diagnostics layer **416**), in monitoring and reporting applications **422**, and/or by system manager **502**, in various embodiments. Such processing circuitry may be located at the building for which the building timeseries data is descriptive (e.g., part of BMS controller **366**, part of a lower level controller, part of an edge device, etc.), may be remote from such a building (e.g., in a server farm, cloud computing system, etc.), or may be distributed over some combination thereof.

(98) At step **802**, building timeseries data is obtained for a time period. The building timeseries data can be obtained from various sensors, equipment, controllers, meters, etc. of a building management system. In some embodiments, the building timeseries data includes one or more virtual points and step **802** includes calculating timeseries values for the one or more virtual points (e.g., based on a combination of measurements/settings/etc., based on estimates, based on modeling, etc.). The building timeseries data can include values for multiple points (e.g., variables, conditions, states, statuses, etc.) for each time step in a time period (e.g., every minute, every 15 minutes, every hour, etc.). Step **802** can include aligning timeseries data to a common sample rate and/or performing other pre-processing steps to improve data quality.

(99) At step **804**, data loss, if any, is detected. Data loss refers to periods where data for a point is not available or not reliable, for example due to loss of communications connectivity, communications errors, loss of power to a sensor, etc. Step **804** can include determining time steps for which data is not available for any of the points of the building timeseries data, for all of the points of the building timeseries data, for a particular point of the building timeseries data, etc. in various embodiments. Detecting data loss may include identifying null values for one or more points for one or more time steps. Detecting data loss may include identifying extreme outlier values indicating sensor malfunction, communications failure, etc. (e.g., values of zero for room temperature typically around 60-80 degrees Fahrenheit). Step **804** can include determining a start time and end time of a period of data loss (e.g., between which data is not available or not reliable).

(100) At step **806**, one or more faults are detected using the building timeseries data. Faults can include various equipment or system failures or degradations, for example relating to failure of equipment to track a setpoint, broken hardware, inefficiencies, and the like. Step **806** can include running rules-based fault detection logic (e.g., comparing point values to a threshold value, testing a statistical property of a set of point values, etc.), machine learning fault detection logic, or other approach to detecting (and, in some embodiments, diagnosing) faults in the building equipment. Step **806** can be provided using the same or similar algorithms and approaches as described above for FDD layer **416** and/or can be performed by FDD layer **416** in some embodiments. Detecting a fault in step **806** can include determining a start time for the fault and, where applicable, an end time for the fault (e.g., between which the fault is occurring, immediately outside of which the fault is not occurring). Step **806** can include suppressing duplicate faults, determining the criticality or severity of a fault, identifying the equipment associated with a fault, or other features relating to fault detection and diagnosis.

(101) At step **808**, subperiods of the time period are classified as normal, faulty, or data loss. A subperiod is classified as a data loss subperiod if data loss occurred for that subperiod, as determined in step **804**. For example, the set of time steps that occur between the start times and end times of occurrences of data loss can be classified as data loss subperiods. A subperiod is classified as a faulty subperiod if a fault was detected for the subperiod in step **806**. For example, the set of time steps that occur between the start times and end times of occurrences of faults can be

classified as faulty subperiods. In some embodiments, a given time step or subperiod can be classified as both a data loss time step or subperiod and a faulty time step or subperiod if the given time step or subperiod meets the criteria of both steps **804** and **806**. Other time steps or subperiods may be classified as one of a data loss time step or subperiod or a faulty time step or subperiod, or may not be classified as either a data loss time step or subperiod or a faulty time step or subperiod. After identification of data loss subperiods and faulty subperiods, the remaining subperiods (e.g., time steps between or outside the data loss subperiods and the fault subperiods) are classified as normal. Other classifications can be used in addition or as alternatives to the normal, faulty, and data loss classification of step **808** in other embodiments.

(102) At step **810**, lines or bars (or other representations) showing the building timeseries data are graphed (plotted, etc.), with the normal subperiods, faulty subperiods, and data loss periods distinguishable based on visual characteristics of the graphed lines or bars. Step **810** can include generating the graphical user interface **600** as shown in FIG. **6**, the graphical user interface **700** as shown in FIGS. **7**A-B, or as adapted to show data for another unit or type of equipment, system, or building. Other interfaces, dashboards, visualizations, etc. are also possible within the scope of step **810**. The visual characteristics which distinguish the normal subperiods, faulty subperiods, and data loss periods can include color, brightness, thickness, translucency, pattern (e.g., solid, dotted, dashed, combination thereof), etc. Such visual characteristics are direct attributes of the plotted lines/bars of data, such that the subperiod classification is presented without addition of extra lines, text, shading, highlighting, etc. which may otherwise clutter the display and/or obstruct views of data of interest.

(103) Process **800** can then proceed to step **812** and/or step **814**. At step **812**, the lines or bars generated in step **810** are presented in a graphical user interface, for example as the graphical user interface **600** illustrated in FIG. **6**. Step **812** can include providing an interactive dashboard to a user via a client device (e.g., laptop computer, smartphone, tablet, virtual reality headset, etc.). In some embodiments, step **812** includes exporting the lines or bars in a .pdf file, as vector graphics (e.g., a scalable vector graphic (SVG) file or other types of path-based graphics or area-based graphics, an image file (e.g., .png, .jpg), or other format. In some embodiments, step **812** includes controlling a printer to print a page showing the lines or bars.

(104) At step **814**, a dynamic audiovisual presentation is generated which includes the graphed lines or bars. The dynamic audiovisual presentation may be similar to the embodiments described with reference to the graphical user interface **700** of FIGS. **7**A-B. In some embodiments, the dynamic audiovisual presentation is populated with desired values of the building timeseries data for the time period and with the lines or bars from step **810**. In some embodiments, plotted lines or bars may be too long to fit on a screen at one time while preserving a scale that provides a meaningful view to a user (longer than a width of the screen). The dynamic audiovisual presentation can be populated with such lines, with the dynamic audiovisual presentation including continuously sliding such lines or bars across a display window so that each portion of the lines or bars is displayed on the screen for a limited time (i.e., the amount of time required for the portion to slide across the width of the screen) and the full lines or bars (e.g., data for the entire time period) are displayed by an end of the dynamic audiovisual presentation. The dynamic audiovisual presentation generated in step **814** may also include an audio track, for example an audio track based on the subperiod classifications from step **808**. For example, the audio track may provide silence during portions of the audiovisual presentation corresponding to normal subperiods and sound (e.g., tones, alarms, beeping, buzzing, notes, voices, etc.) during portions of the audiovisual presentation corresponding to faulty subperiods and/or data loss subperiods (or vice versa). Generating the dynamic audiovisual presentation in step **814** can include selecting a characteristic of the sound for the audio track (e.g., volume, pitch, tone, timbre, etc.) based on a type, severity, or other trait of a corresponding fault.

(105) At step **816**, the dynamic audiovisual presentation is played (e.g., on client device). Step **816**

can including continuously advancing the graphed lines and/or bars across a display screen, for example such that the time steps of the time period for which building data was obtained in step **802** slide continuously across the display screen. A full set of historical data is thereby displayed over time as the dynamic audiovisual presentation is played. Step **816** may also include playing an audio track of the dynamic audiovisual presentation in sync with the visual track (i.e., synced with movement of the graph lines and/or bars), for example so that sound is emitted according to the audio track at the same time that a fault subperiod (or other feature of a graph) passes through a particular point or region on the display screen (e.g., a middle of the screen, a right edge of the screen, a left edge of the screen, a vertical line designated on the screen, etc.). One or more speakers (speakers, headphones, etc.) can be controlled in step **816** to emit the sound according to the audio track of the dynamic audiovisual presentation. Rich building timeseries data and status information (e.g., normal, data loss, fault, etc.) is thereby communicated to a user in an easy-to-understand manner which does not require scrolling, navigating, zooming in, zooming out, etc. to access all available data.
(106) In some embodiments, the various timeseries presented via the graphical user interfaces **600** and **700** are generated and available prior to receiving a request to generate the graphical user interfaces **600** or **700**. For example, the timeseries may be stored in memory locally or remotely and retrieved when a user requests to display the graphical user interface **600** or play the graphical user interface **700**. The stored timeseries can include, for example, a fault detection timeseries that indicates whether a particular fault exists or does not exist at each time step of the fault detection timeseries, or a data loss timeseries that indicates whether data loss is present or absent at each time step of the data loss timeseries. Alternatively, one or more of the timeseries presented via the graphical user interfaces **600** and **700** can be generated in real-time during playback or in response to a request to present the graphical user interfaces **600** and **700**. For example, the system or component that generates the graphical user interfaces **600** and **700** can perform real-time fault detection or data loss detection on the timeseries data to generate supplemental fault detection timeseries or data loss timeseries in response to a request to generate the graphical user interfaces **600** and **700**.
(107) In some embodiments, the audiovisual data presented via graphical user interfaces **600** and **700** can be pre-rendered or pre-generated prior to receiving a request to generate or play the graphical user interfaces **600** and **700**. For example, the complete visual presentation of the line graphs and bars shown in the graphical user interface **700** can be pre-rendered (e.g., as an image or vector graphic) prior to beginning playback and the time window can slide forward in time over the pre-rendered visual data during playback. Alternatively, some or all of the audiovisual presentation can be generated or rendered in real-time during playback, for example, as the timeseries data slides into the data window during playback.
(108) The features shown in FIGS. **6-8** and described herein advantageously present fault information and building data in a manner that allows building operations to be assessed and adjusted to improve equipment and system performance. For example, insights can be presented in an understandable manner which helps drive reduction in faults, reduction in equipment downtime, improved maintenance efficiency, reduced energy consumption, etc. Further, the approaches and interfaces described herein solve previous challenges associated with cluttered and obstructed views and with requiring users to perform undesirable amounts of page navigation, scrolling, and zooming, for example by streamlining views, streaming visual presentations, and emitting sound in sync with such data presentation. The teachings herein thus both solve technical challenges associated with surfacing building timeseries data to human users and provide for improved performance of building equipment.
Configuration of Exemplary Embodiments
(109) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been

described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

(110) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products including machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(111) Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule Based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision step.

Claims

1. A method for a building management system, comprising: detecting a fault from timeseries building data; classifying a time period of the timeseries building data as a faulty time period responsive to detecting the fault; playing an audiovisual presentation of the fault and the timeseries building data via a screen and a speaker by: visually advancing through the timeseries building data in a chronological order such that at least a normal time period and the faulty time period are presented at different times, wherein the timeseries building data is presented with different coloring, dashing, or line type in the normal time period as compared to the faulty time period, wherein visually advancing the timeseries building data comprises moving a chart of the timeseries building data across the screen, and wherein a width of the chart exceeds a width of the screen; and emitting an audio tone when the timeseries building data for the faulty time period is presented.

2. The method of claim 1, comprising determining a quality of the tone based on a type or severity of the fault.
3. The method of claim 2, wherein the quality of the tone comprises a pitch or volume of the tone.
4. The method of claim 1, wherein a pitch of the tone is a function of an amount of deviation of a point of the timeseries building data from a target or threshold value for the point.
5. The method of claim 1, further comprising stopping the tone when the timeseries building data for the faulty time period is no longer presented.
6. One or more non-transitory computer-readable media storing program instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: detecting a fault from timeseries building data; classifying a time period of the timeseries building data as a faulty time period responsive to detecting the fault; playing, by controlling a screen and a speaker, an audiovisual presentation of the fault and the timeseries building data by: visually advancing through the timeseries building data in a chronological order, wherein the timeseries building data is presented with different coloring, dashing, or line type in the normal time period as compared to the faulty time period, wherein visually advancing the timeseries building data comprises moving a chart of the timeseries building data across the screen, and wherein a width of the chart exceeds a width of the screen; and emitting an audio tone when the timeseries building data for the faulty time period is presented.
7. The one or more non-transitory computer-readable media of claim 6, the operations further comprising determining a pitch of the tone based on a type or severity of the fault.
8. The one or more non-transitory computer-readable media of claim 7, wherein the pitch of the tone is a function of an amount of deviation of a point of the timeseries building data from a threshold or target value for the point.
9. The one or more non-transitory computer-readable media of claim 6, the operations further comprising stopping the tone when the timeseries building data for the faulty time series is advanced such that the timeseries building data is no longer presented.
10. A system, comprising: one or more processors; and one or more non-transitory computer-readable media storing program instructions that, when executed by the one or more processors, cause the one or more processors to perform operations comprising: detecting a fault from timeseries building data; classifying a time period of the timeseries building data as a faulty time period responsive to detecting the fault; playing, by controlling a screen and a speaker, an audiovisual presentation of the fault and the timeseries building data by: visually advancing through the timeseries building data in a chronological order, wherein the timeseries building data is presented with different coloring, dashing, or line type in the normal time period as compared to the faulty time period, wherein visually advancing the timeseries building data comprises moving a chart of the timeseries building data across the screen, and wherein a width of the chart exceeds a width of the screen; and emitting an audio tone when the timeseries building data for the faulty time period is presented.
11. The system of claim 10, the operations further comprising determining a pitch of the tone based on a type or severity of the fault.
12. The system of claim 11, wherein the pitch of the tone is a function of an amount of deviation of a point of the timeseries building data from a threshold or target value for the point.
13. The system of claim 10, the operations further comprising stopping the tone when the timeseries building data for the faulty time series is advanced such that the timeseries building data is no longer presented.